AUGUST 2026

2026-08-01 | 07:00 PM – 10:00 PM:

OBJECTIVE: ***Expanding my understanding of Parkinson's disease***
NEXT STEP: ***Find a direction for my research and look into each symptom***

Studied the symptoms of Parkinson's disease, particularly bradykinesia, freezing of gait, paradoxical kinesia, anosmia, and RBD. The role of dopamine as both a neurotransmitter and a neurohormone with peripheral function was researched much more in depth, with my notes covering its nature beyond acting as the primary neurochemical in motivation.

2026-08-02 | 01:00 PM – 03:00 PM:

OBJECTIVE: ***Researching anosmia and finding a thesis***
NEXT STEP: ***Narrow down my thesis***

After writing a short note on anosmia in my research tab for Parkinson's disease, I expanded those notes in a new tab on my research doc. I covered its definition, statistics in the American population, and its difficult detection, caused by a reliance on self-reporting and several possible factors that come with cases of olfactory dysfunction. Furthermore, I found a general direction for my research. A big picture inquiry; "How can the occurence of non-motor conditions, primarily anosmia and REM sleep behavior disorder (RBD) be studied and used for the screening and early detection of Parkinson's disease before the onset of motor dysfunction?" Alongside a comparative inquiry; "How can non-motor biomarkers shared by synucleinopathic disorders such as Alzheimer's and Dementia with Lewy Bodies, be studied for early detection before the later stages of these conditions?"

Research in this phase primarily cited the ScienceDirect journal and the National Institutes of Health.

2026-08-03 | 06:00 PM – 07:00 PM:

OBJECTIVE: ***Researching muscle atonia***
NEXT STEP: ***Find rich datasets and parse them***

Although anosmia remained in my general thesis at this point, I met a dead end on data because anosmia, the complete loss of one's sense of smell, is very rare, with few detailed statistics available for the Canadian population, alongside the fact that olfactory dysfunction can be a product of several factors, such as medication, aging, injury, et cetera. I did more research into

RBD, starting with muscle atonia. I dove more into Rapid Eye Movement, particularly one of its foundation elements, muscle atonia, a temporary paralysis induced during this phase of sleep to stop physical dream enactment. My primary resource here was the National Institutes of Health.

2026-08-04 — 2026-08-06:

OBJECTIVE: *Finding rich, rigorous datasets.*
NEXT STEP: Log into Google Colab and Learn Pandas To Parse My Data

This period was spent trying to look for a dataset. I was originally going for the Michael J. Fox Foundation's Parkinson's Precision Medicine Initiative datasets. However, there proved to be a barrier to entry as it is primarily designed for institutional researchers, and although I understood it was to ensure the privacy of every patient profiled in their data, I did not feel comfortable giving away personal information such as my school, home address, and ZIP code. The next move I made was searching through Kaggle. I found several projects covering symptoms and popular spaces in the research of Parkinson's disease, such as the analysis of voice recordings, handwriting, freezing of gait, drawings, recorded tremor data, and much more. I chose not to go with these projects because they are covering late-stage conditions, and my research was about differentiating itself by focusing on prodromal markers, such as RBD.

I didn't want to give up, so I looked more into Kaggle and eventually found a dataset, by Rabie El Kahoura, of what seemed to be over 2,105 rigorously studied patients. This set covered several details spanning blood sugar and body mass index, to alcohol consumption and medical history. As exhaustive as it was, this did not make the cut as I discovered it was a synthetic set of artificial profiles, explaining how so much data was recorded on so many profiles without major privacy and ethical concerns. Eventually, I found a set of 130 profiles by the UC Irvine Machine Learning Repository, exhaustive accelerometer data recording the movement of one patient, the ALAMEDA project by Konstantina-Maria Giannakopoulou et al, and the PADS (Parkinson's Disease Smartwatch) set by MIT's PhysioNet site, covering 469 profiles, which I failed to download due to storage issues.

2026-08-07 | 02:00 PM – 08:00 PM:

OBJECTIVE: *Learn the basics of Pandas and write observation notes for the UCI set*
NEXT STEP: *Filter through the set for specific profiles and features*

I uploaded the UCI set's CSV file to my Google Drive and connected my account to Google Colab to access the set. I started a one-hour YouTube tutorial to quickly learn the basics of Pandas, and I created the 'Observations' tab in my research doc to log down observation notes. With over 130 profiles to track across several categories, a mix of Parkinson's disease patients, RBD patients, and healthy individuals, there is a fair bit of work to do. I broke the set into subcohorts and began with the "Young Cohort", consisting of 13 individuals below the age of 50. What caught my eye most, though, were the four RBD patients in this profile. RBD is a very strong indicator of a future

synucleinopathy, including Parkinson's disease, with an 80% conversion to a disorder in that family. However, that is only true for idiopathic RBD. These are isolated cases of RBD, and they hold the 80% conversion figure. The same cannot be said about their counterpart, medication-induced RBD. Three of the four RBD patients in the Young Cohort were on medications known to induce RBD, such as Anafranil and Cipralex. Due to this, I had to exclude them from most future analyses, leaving me with only one confirmed case of idiopathic RBD by an individual below 50 across 130 profiles.

Alongside this, I recorded information such as the Entropy of Speech Timing of the subcohort and their mean age.

```
# For the official data and CSV, visit the link on this repository's home page
# I will be using my Google Drive to access all imported data for this project
# Links and files will be posted for access in the 'Resources' folder

import pandas as pd

from google.colab import drive
drive.mount('/content/drive')

uciRead = pd.read_csv("/content/drive/MyDrive/Parkinson's Research/Datasets/uci-pd-data.csv", index_col = "Unnamed: 0")

uciRead["Demographic information"] = pd.to_numeric(uciRead["Demographic information"], errors="coerce") # The "Age (Years)" subcolumn causes Pandas to interpret the greate

youngProfiles = uciRead[uciRead["Demographic information  "] <= 50] # Give me all healthy and diagnosed profiles below the age of 50
print(youngProfiles.to_string())
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
      Demographic information Demographic information.1 Clinical information Clinical information.1 Clinical information.2       Medication Medication.1 Medication.2  Med
PD12                    37.0                         F                   No                    32                    5               No           No           No
PD13                    43.0                         M                   No                    37                    6               No           No           No
PD19                    34.0                         M                   No                    30                    4               No           No           No
RBD23                   46.0                         M                   No                    36                   10     Yes (Zoloft)           No           No
RBD27                   48.0                         M                   No                    46                    2               No           No           No
RBD28                   47.0                         M                   No                    40                    7  Yes (Anafranil)           No           No
RBD50                   40.0                         F                   No                    33                    7   Yes (Cipralex)           No           No  Yes
HC02                    45.0                         M                    -                     -                    -               No           No           No
HC07                    45.0                         M                    -                     -                    -               No           No           No
HC09                    45.0                         M                    -                     -                    -               No           No           No
HC21                    40.0                         F                    -                     -                    -               No           No           No
HC45                    46.0                         M                    -                     -                    -               No           No           No
HC49                    44.0                         M                    -                     -                    -               No           No           No
```

2026-08-10 | 06:00 PM - 07:00PM:

OBJECTIVE: *Record new observation notes*
NEXT STEP: *Research alpha-synuclein*

As I learned Pandas, I recorded more information on the profiles, such as the age of disease onset, for the three PD patients and one iRBD (idiopathic RBD) patient and duration from first symptoms. I also created my Vocabulary page to define terms and metrics I would be using in my research, such as 'entropy of speech timing', 'Hoehn & Yahr scale', and 'UPDRS III'.

2026-08-12 | 06:00 PM - 07:00 PM:

OBJECTIVE: *Research alpha-synuclein*
NEXT STEP: *Look into the motor examination of the patients*

I did more research into what alpha-synuclein was. I noted down its nature as a presynaptic neuronal protein and its relationship in brain tissue as a major neurochemical in the 'sending' part of communication between nerve cells through axon terminals. My primary sources here were the National Institutes of Health and ScienceDirect, the latter from which I retrieved a diagram of alpha-synuclein in action.

2026-08-13 | 02:00 PM - 04:00 PM:

OBJECTIVE: Mapping out the UPDRS III motor examination
NEXT STEP: *Creating the first graphs for the speech examination scores*

I looked over the Hoehn & Yahr Scale of the three PD patients in the Young Cohort, determining that all three were facing bilateral impairment, with PD12 and PD13 both at Stage 2, and PD19 at Stage 3. Furthermore, I mapped out the patients' scores (PD12, PD13, PD19, RBD27) on 'Speech', 'Facial Expression', and 'Tremor at Rest - Head'. The most interesting of these details is the first, Speech. All patients had a score of 0 in this category, indicating completely normal speech. I expect Parkinson's disease, especially in Stages 2 and beyond on the Hoehn & Yahr scale, to come with some notable irregularity in speech. All patients in the Young Cohort scoring 0 on Speech, in combination with entropy of speech timing nearly uniform with that of the healthy controls, was quite odd for me, but it gave me a more specific angle to explore.

Besides speech, I discovered the PD19 had a case of hypomimia, difficulty in forming simple facial expressions that show the one's emotions, such as smiles. Hypomimia was not a major detail here as it does fall in line with established trends concerning this patient's progression. The four other patients scored a 0 in this category. Finally, I discovered PD19 scored a 1 in the 'Tremor at Rest - Head' category, indicating infrequent shaking (rest tremors) of the temple. All other patients scored a 0 here, however, rest tremors are not a major detail here as they usually come with the later stages of Parkinson's disease.

The goal for the next day is to learn Matplotlib so I can begin plotting simple graphs for the cohort's speech examination scores.

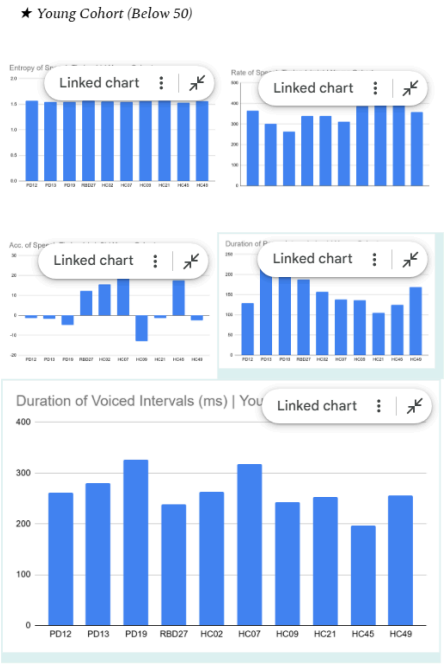
2026-08-14 | 10:00 AM - 09:00 PM:

OBJECTIVE: Graphing in Google Sheets and learning the basics of Matplotlib
NEXT STEP: *Map out the rest of the speech examination in Matplotlib*

I did not have any prior experience with Matplotlib at this point, so to make the entry easier, I simply used Google Sheets. With my data ready in Pandas, I created a Google Sheet and visualized it using basic bar graphs. I modelled the entropy of speech timing, rate of speech timing, acceleration of speech timing, duration of pause intervals, and duration of voiced intervals. I then wrote down my observation notes. As I noticed before, when calculating the average, the entropy of speech timing was near-uniform across the Young Cohort. I recorded inconsistencies between

the rate of speech timing and the acceleration of it, where the three PD patients and RBD27 recorded a 0 in Speech back in the UPDRS Part III motor examination. All PD patients showed a decline in the rate of speech timing, relative to the rest of the cohort, while all three had scores in the negatives on the acceleration of speech timing metric.

However, this was a cohort of 10 people, including only six actual patients, so I could not draw any meaningful conclusions from these findings just yet. I began a one-hour tutorial on YouTube on Matplotlib, and once I got a handle on the basics, I created a line graph to visualize the acceleration of speech timing, though I had to get used to determining significant details in these graphs compared to if they had been represented by bars.



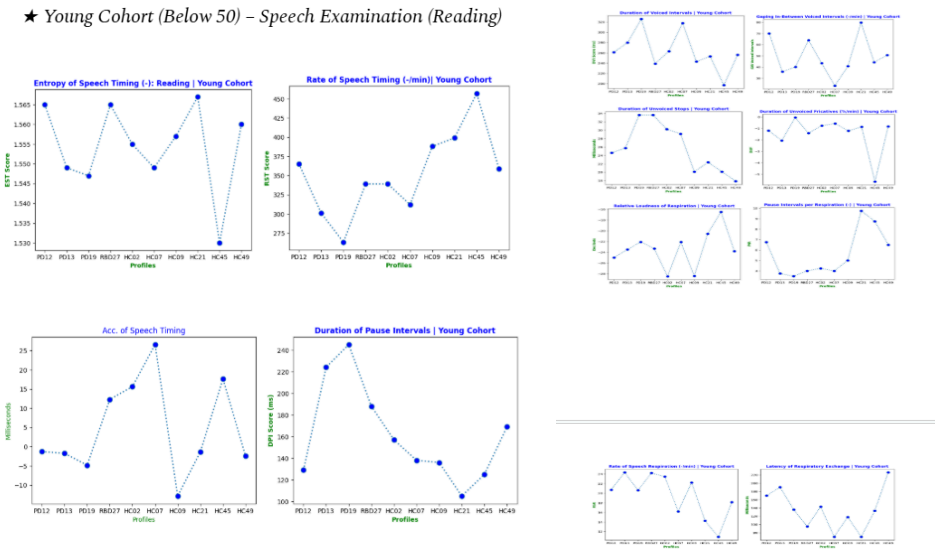


2026-08-15 | 03:00 PM - 08:00 PM:

OBJECTIVE: Graph the speech examination figures for the Reading category
NEXT STEP: Graph the speech examination figures for the Monologue category

Feeling confident, I went over each other category in the "Speaking Task of Reading Passage" half, from the entropy of speech timing and rate of speech timing, to the rate of speech respiration and latency of respiratory exchange. This was followed by more observation notes, where I noticed that the Health Controls had relatively short pauses during speech, whereas PD13, PD19, and RBD27 showed the longest in the group, all above 180 millidseconds. While 180 milliseconds is only a fraction of a second, its significance lies in the context of these patients scoring 0 in Speech on the motor examination. I knew speech was not a monolith, but seeing these details and quirks about the speech of these individuals that aren't detectable by clinical audits relying on the ear opened my eyes to the meaning of that concept.

For perspective, the rest of the cohort all had scores below 180 milliseconds, and that range is skewed by HC07 and HC49, who had scores above 160 milliseconds, with everyone else below 140 milliseconds. These inconsistencies gave me much greater clarity on what direction to take the project in. As such, I finally came up with a rough thesis that would evolve and narrow down as the project went on; "Quantifying subclinical non-motor dysfunction in the prodromal window of Parkinson's disease to reduce reliance on subjective clinical audits".

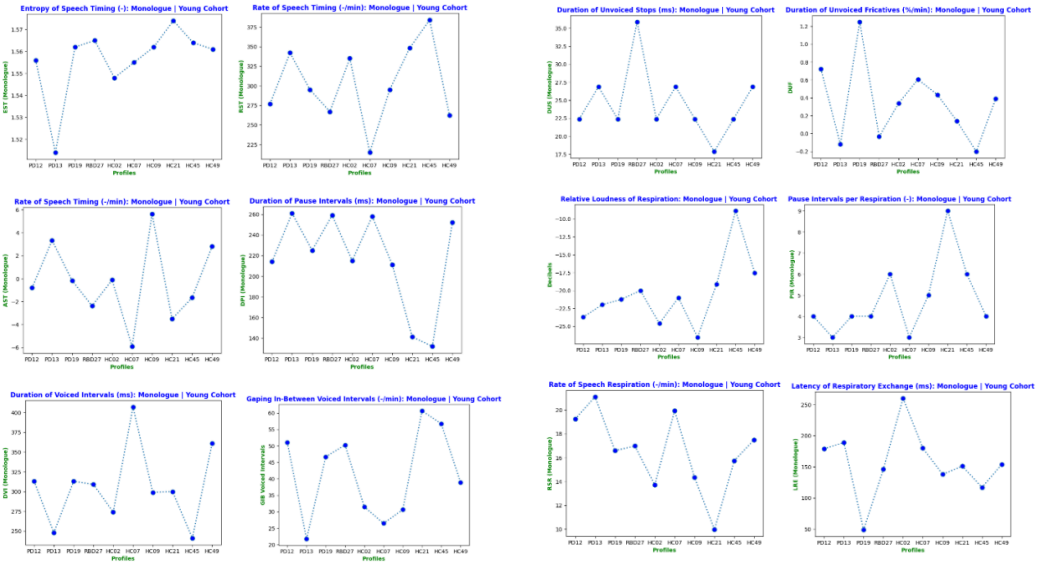


2026-08-17 | 11:00 AM - 03:00 PM:

OBJECTIVE: Graph the speech examination figures for the Monologue category
NEXT STEP: Begin organizing data for the Middle Cohort (50 to 64)

I graphed the rest of the speech examination for the last "Speaking Task of Monologue" half. I began to question whether each metric was actually useful for my analysis, so I decided to hold off on comparing these graphs to their counterparts in the Reading half. I moved on to defining who would be in the Middle Cohort and began exploring how to create more complex graphs in Matplotlib, such as boxplots, violin plots, et cetera, that would show significance and distribution more accurately, though I hadn't learned how to do these yet in Matplotlib.

★ *Young Cohort (Below 50) – Speech Examination (Monologue)*



2026-08-18 | 11:00 AM - 03:00 PM:

OBJECTIVE: Clean up Pandas code
NEXT STEP: Model the Middle Cohort

As expected of a dataset covering 130 individuals, the Middle Cohort, of profiles above the age of 50 and below 64, and the Older Cohort, anyone 65 years of age and older, were much larger than the 10-person Young Cohort (after removing three cases of medication-induced RBD). To make this easier, I wrote code for my program to ask the user for "commands". These commands would be "Young Cohort", "Middle Cohort", and "Older Cohort", and they would tell Pandas to print me their corresponding cohorts. I began planning how I'd sort through so many profiles and landed on setting up their Reading and Monologue graphs side by side.



```
# Commands to make searching for cohorts easier

# print(youngProfiles.to_string())

command = input("Command: ")

if command == "Young Cohort":
  try:
    print(youngProfiles.to_string())
  except KeyError:
    print(f"{command} does not exist")

elif command == "Middle Cohort":
  try:
    print(middleProfiles.to_string())
  except KeyError:
    print(f"{command} does not exist")

elif command == "Older Cohort":
  try:
    print(olderProfiles.to_string())
  except KeyError:
    print(f"{command} does not exist")
```

2026-08-19 | 05:00 PM - 11:00 PM:

OBJECTIVE: Organize the Middle Cohort and download the PhysioNet PADS Set
NEXT STEP: Graph out Speech Examination (Reading & Monologue)

Using the commands I wrote in the day before, I set up the Middle Cohort to look through, then I ran into a conundrum. Some RBD patients were on medications such as Rivotril and Xanax, which have a notable impact on movement, so excluding them was an easy task, giving me a total of 14 cases of idiopathic RBD, which is much more statistically significant than the only iRBD patient I had to work with in the Young Cohort. But I was a bit unsure of what to do when it came to the PD patients. Some had been on medication not specified but indicated in the CSV file, but because they had already been formally diagnosed with Parkinson's disease, I was unsure of whether these medications would really impact scores on the UPDRS Part III examination. At the same time, I did not want to exclude these profile as I would be left with only six.

After some time, I decided that any medication at all poses a risk of exaggerating pre-existing motor conditions, whether involving tremors, bradykinesia, or the basal ganglia that facilitate speech. I couldn't manually skip profiles on medication as I could with the Young Cohort of 13 individuals, so I wrote some code so Pandas could exclude any profile that showed positive for any medication, specified or not.

Furthermore, I finally cleared out enough space on my computer to download the total PhysioNET PADS (Parkinson's Disease Smartwatch) dataset, so now I had several JSON files sitting on my computer, collectively representing 469 profiles. This was a huge, exciting success, but I did not know how to parse the data yet, so I let it on the back burner while I worked through the Middle Cohort.

```
uciRead["Demographic information"] = pd.to_numeric(uciRead["Demographic information"], errors="coerce") # The "Age (Years)" subcolumn causes Pandas to interpret the grea
uciRead["Medication.5"] = pd.to_numeric(uciRead["Medication.5"], errors="coerce")

youngProfiles = uciRead[uciRead["Demographic information"] < 50] # Give me all healthy and diagnosed profiles below the age of 50
middleProfiles = uciRead.loc[(uciRead["Demographic information"] >= 50) & (uciRead["Demographic information"] < 65) & (uciRead["Medication"] == "No") & (uciRead["Medicat
olderProfiles = uciRead[uciRead["Demographic information"] >= 65]
```

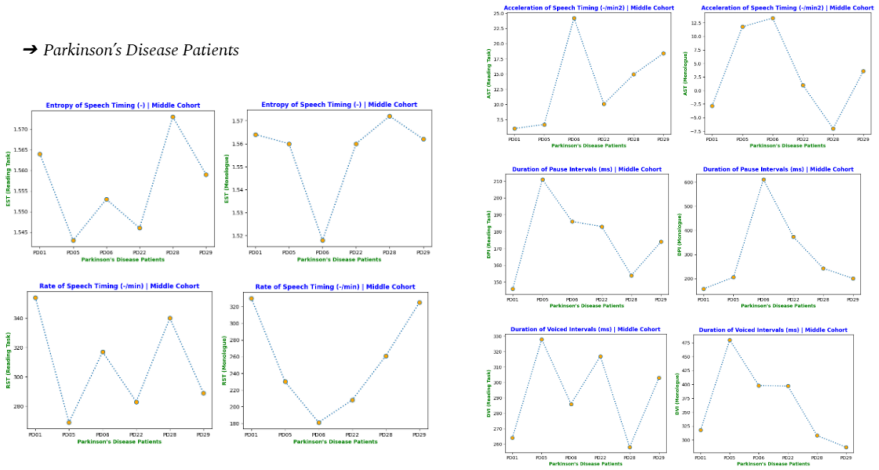
2026-08-20 | 07:00 PM - 09:00 PM:

OBJECTIVE: Graph out Speech Examination for the PD patients, side by side.
NEXT STEP: Graph out Speech Examination for the PD and iRBD patients

This was a relatively lighter workday. I graphed out the entropy of speech timing, rate of speech timing, acceleration of speech timing, duration of pause intervals, and duration of voiced intervals for the Reading and Monologue halves of the speech examination. I decided to hold off on creating more advanced graphs for a bit until I was ready, and this day was more focused on getting the graphs down and saving note-taking for later.

★ *Middle Cohort (50 to 64) – Speech Examination*

➔ *Parkinson's Disease Patients*

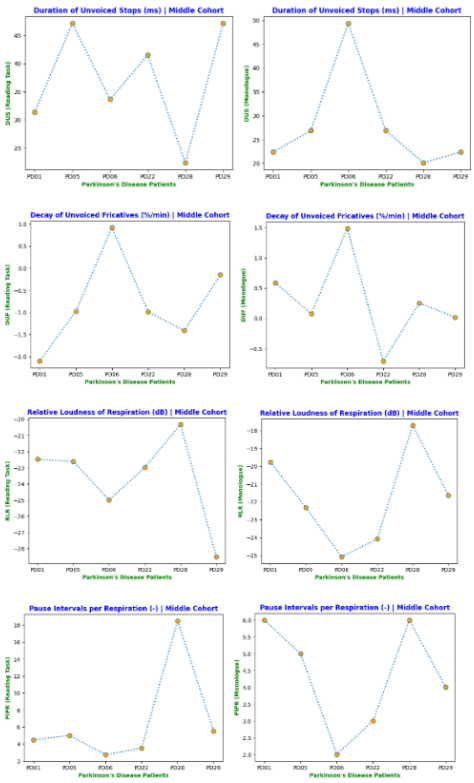


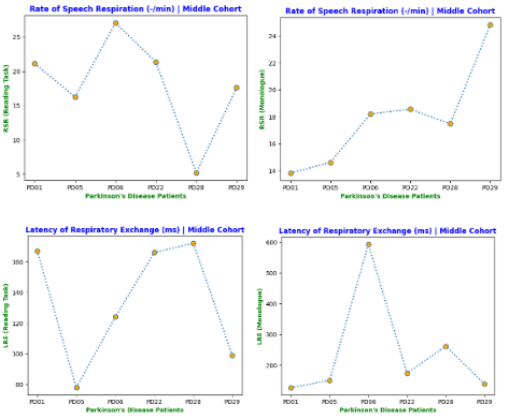
2026-08-21 | 07:00 PM - 09:00 PM:

OBJECTIVE: Finishing graphing for the PD patients and get started on the iRBD group
NEXT STEP: Finishing graphing for the iRBD patients

This was another lighter, straightforward day. I finished graphing speech examination for the PD patients and graphed out the entropy, rate, and acceleration of speech timing for the iRBD group. Besides finishing graphing for this group, two buzzing items to mark off my list are exploring the PhysioNET data to get much better clarity on how it is structured and how I can read it, and finally

setting up the GitHub for my project so I can track and log changes to my code more simply and efficiently.






```
[27]
✓ 0s
import matplotlib.pyplot as plt
import numpy as np

# PROFILES

pdProfiles = ["PD01", "PD05", "PD06", "PD22", "PD28", "PD29"] # PARKINSON'S DISEASE PATIENTS (NO MEDICATION)
rbdProfiles = ["RBD02", "RBD05", "RBD07", "RBD12", "RBD16", "RBD18", "RBD19", "RBD22", "RBD29", "RBD35", "RBD40", "RBD43", "RBD45", "RBD49"]
healthyControls = [] # HEALTHY INDIVIDUALS (NO MEDICATION)

# Speech Examination Figures (PD PATIENTS)

# ENTROPY OF SPEECH TIMING
estScoreReading = [1.564, 1.543, 1.553, 1.546, 1.573, 1.559]
estScoreMonologue = [1.564, 1.56, 1.518, 1.56, 1.572, 1.562]

# RATE OF SPEECH TIMING
rstScoreReading = [354, 269, 317, 283, 340, 289]
rstScoreMonologue = [330, 230, 181, 208, 261, 325]

# ACCELERATION OF SPEECH TIMING
astScoreReading = [6.05, 6.72, 24.19, 10.09, 14.94, 18.4]
astScoreMonologue = [-2.82, 11.77, 13.38, 1, -7.02, 3.57]

# DURATION OF PAUSE INTERVALS
dpiScoreReading = [146, 211, 186, 183, 154, 174]
dpiScoreMonologue = [158, 206, 611, 374, 243, 201]

# DURATION OF VOICED INTERVALS
dviScoreReading = [264, 328, 286, 317, 258, 303]
dviScoreMonologue = [318, 480, 398, 397, 308, 287]
```

2026-08-22 | 04:00 PM - 05:00 PM:

OBJECTIVE: Begin graphing Speech Examination scores for the iRBD group
NEXT STEP: Finishing graphing these figures.

The iRBD demographic of the Middle Cohort is composed of 14 patients diagnosed with RBD and on no medication. This is obviously much more statistically powerful than the one iRBD patient I had in the Young Cohort, and with varying scores across 24 categories, it was quite tedious running through this group. On this day, I entered the scores for the entropy, rate, and acceleration

of speech timing metrics across both the reading and monologue halves of the examination in a new section of my Google Colab file.

```
# Speech Examination Figures (iRBD PATIENTS)

estScoreReading2 = [1.564, 1.551, 1.541, 1.56, 1.564, 1.552, 1.559, 1.565, 1.468, 1.534, 1.559, 1.556, 1.485, 1.537]
estScoreMonologue2 = [1.568, 1.555, 1.56, 1.535, 1.541, 1.563, 1.555, 1.543, 1.536, 1.544, 1.556, 1.567, 1.547, 1.55]

rstScoreReading2 = [359, 358, 270, 285, 278, 342, 361, 354, 293, 337, 296, 294, 317, 297]
rstScoreMonologue2 = [286, 273, 288, 234, 247, 286, 273, 273, 246, 259, 269, 307, 255, 266]

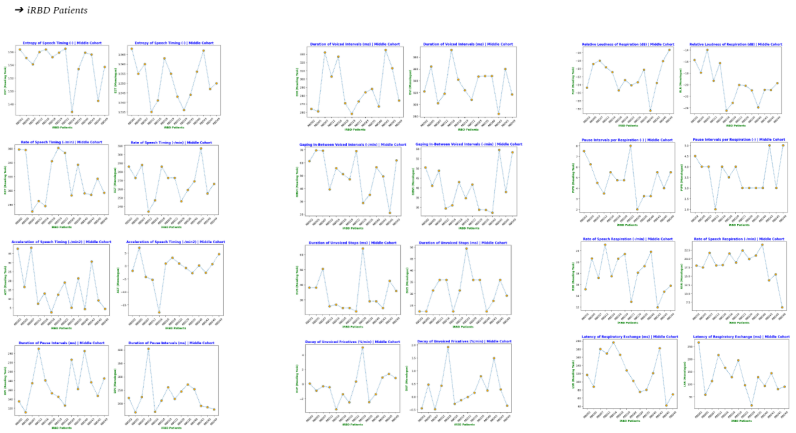
astScoreReading2 = [37.62, 16.54, 38.15, 7.3, 13, 2.54, 12.31, 19.01, 5.07, 21.35, 4.4, 30.59, 9.15, 4.46]
astScoreMonologue2 = [-1.89, 7.01, -4.21, -5.3, -17.87, 0.97, 3.21, 0.93, -0.78, -2.74, 0.21, -2.59, 0.7, 4.61]
```

2026-08-24 | 06:00 PM - 09:00 PM:

OBJECTIVE: Complete graphing for the iRBD cohort
NEXT STEP: Explore and understand the PhysioNet PADS dataset

I completed every graph for every category in the speech examination for the iRBD group, and this took a little longer than I thought it would, as some of these graphs did get mixed up as I looked over the examination. While graphing the scores for the Healthy Controls feels more like the logical next step, I have to admit that creating these graphs is quite repetitive, and I feel I am not

generating anything "novel" right now. As such, I will look through the 469 profiles presented in the PADS data, and will pause serious research in preparation for the grade 11 school year. I am also looking into Scikit (also Scipy, "SKLearn") to begin running statistical tests on the data I've logged in so far and identify more advanced features, such as distributions and deviations. Furthermore, the GitHub repository for this project has finally been figured out and created.



2026-08-31 | 07:00 PM - 08:00 PM:

OBJECTIVE: Outline September-October Roadmap
NEXT STEP: Begin graphing Speech Examination scores for the Healthy Controls

I outlined the roadmap for September 1st to October 31st for my research, including objectives such as learning Scikit ("SKLearn"), processing the PhysioNet PADS dataset, fleshing out my Vocabulary page, et cetera.